Pavel Makhov

Staff Software Developer // GenAI Platform, Java

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Career

Lightspeed Commerce

Montréal, Canada
Aug 2021 – Present · 5y 3m

Staff Software Developer — Jan 2024 – Present · 2y 10m

As part of the AI Platform squad, I architect and build an agentic AI platform that helps teams deliver production-grade generative features from design and implementation to rollout and operations.

- Built an event-driven framework for AI agents using Java, Spring AI, Kafka, Kubernetes, and DynamoDB.
- Built an AI-assisted workflow that imports existing restaurant menus from photos or PDF scans, converting them into structured menu data.

Senior Software Developer — Aug 2021 – Dec 2023 · 2y 5m

- Drove the decomposition of a monolithic application into domain-aligned Spring Boot microservices to improve scalability and team autonomy.
- Led the technical design and delivery of a platform-wide authorization system based on the Attribute-Based Access Control (ABAC) model.
- Shaped the architecture and design of new APIs and surrounding tooling, advocating design-first API development and modern backend best practices.
- Built a native macOS menubar app that helps engineers manage internal Kubernetes clusters and reproducible development environments.
- Created OpenCode plugins and internal tools that automate repetitive engineering tasks.
- Provided 3rd line support and took part in on-call rotations, postmortems, and resolution of high-impact production incidents, including follow-up improvements.
- Partnered with product and engineering stakeholders in discovery and design sessions to define AI feature requirements, delivery plans, and integration constraints.

Solution Media 360

Montréal, Canada
Jan 2021 – Aug 2021 · 8m

Senior Software Developer

Designed and developed Kotlin-based backend services for a leading Canadian provider of innovative technology solutions for the automotive dealership market. Collaborated closely with cross-functional teams to deliver robust and scalable solutions.

- Implemented an edge-level authorization mechanism for a set of microservices.
- Provided 3rd line application support.
- Onboarded and mentored new team members.

Behavox

Montréal, Canada
Jan 2020 – Jan 2021 · 1y 1m

Java Developer

Worked in a product team building compliance management software solutions, developing data-intensive applications using Java, Kotlin, HBase, and MySQL.

- Led technical design, implementation, and delivery of complex data-intensive applications.

TouchTunes / OctaveGroup

Montréal, Canada
Jun 2018 – Dec 2019 · 1y 7m

Backend Developer

Worked as a Java Backend Developer in the product team for the largest in-venue interactive music and entertainment platform, deployed in over 75,000 bars and restaurants across North America and Europe. Built services in a microservice architecture, with Spring Boot-based applications deployed to AWS in Docker containers managed by Mesos/Marathon. Technology stack: Java 11, Spring Boot, Reactor, Cucumber, MongoDB.

- Implemented backend integration for Apple Music For Business.
- Developed the Access Control System, a set of microservices implementing the ABAC authorization model.
- Developed multiple payment integrations with Worldpay/Vantiv, PayPal, Apple Pay, and Google Pay.

Intelerad

Montréal, Canada
Jan 2018 – May 2018 · 5m

Full Stack Java Developer
Contributed to development of PACS (Picture Archiving and Communication System), focusing on Tech Portal, a customization tool that allows technologists to create multiple worklists based on criteria such as exam status, date range, modality, organizations, locations, and rooms. Tech stack: GWT, CSS/SCSS, AngularJS.

Savoir-Faire Linux

Montréal, Canada
Aug 2016 – Dec 2017 · 1y 5m

Java Developer / Consultant
Successfully delivered multiple consulting projects, collaborating with diverse clients to build high-quality solutions. Provided Java expertise across different stages of project lifecycles.

- Developed a high-performance, highly available Music REST API.
- Developed a real-time data processing application based on Apache Mahout and Apache Kafka.
- Developed a WebRTC Liferay portlet and Content Classifier portlet (based on DL4J library).
- Contributed to building a data pipeline for an ML project.

CERN

Geneva, Switzerland
Oct 2012 – May 2016 · 3y 8m

Java Developer
Contributed to development of the business-critical eFiles document management system based on Alfresco, an open-source market leader in records management. Extended and customized the Alfresco platform, and worked on EDH (Electronic Document Handling), a system with complex workflows covering key business processes at CERN, developing new EDH documents and improving existing functionality.

- Integrated Alfresco with AppDynamics, an application performance management system.
- Developed a web application for 2nd/3rd-line support of eFiles. Technology stack included Spring Boot, AngularJS, and Spring Data REST.

Technical skills

Key competencies:

- Java / Kotlin / Lua / Swift
- Spring / Reactor / Hibernate / JUnit / Mockito / Spock
- Oracle, MySQL / Mongo, Redis, DynamoDB, Couchbase
- Datadog / Dynatrace / Sumologic
- TDD / BDD (Cucumber, Spock)
- Google Cloud (Vertex AI) / Azure AI / Kubernetes / Docker
- REST / Design-first APIs / SDK integrations
- Kafka / RabbitMQ / Event-driven architectures
- Scrum / Agile / Pair programming

Talks

API Development: Shifting from Code-First to Design-First

How moving dozens of microservices to design-first APIs improved API quality, covering collaborative OpenAPI workflows, automated validation and governance that prevent breaking changes.

- Montreal JUG — Montréal Java User Group community meetup — Jan 2026
- ConFoo Conference — Multi-track developer conference in Montréal, 190+ sessions — Feb 2026

Open Source Projects

Awesome WM widgets
Collection of widgets for Awesome WM.
Lua

pullBar
macOS application to show GitHub PRs.
Swift

Jekyll Clean Dark
Theme for Jekyll.
HTML

Jekyll CV Crafter
Jekyll-powered CV generator.
HTML

Education

National Research University of Electronic Technology

Master of Computer Science (2007 – 2012)
https://eng.miet.ru/, Moscow, Russia